NAME ……..…PERSONAL NO…….

P553/2

BIOLOGY

Paper 2

July/August 2025

2½ hours

MUKONO EXAMINATION COUNCIL 2025

Uganda Certificate of Education

BIOLOGY

PRACTICAL

Paper 2

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two** examination items. Answer **all** the items in the spaces provided.

Drawings should be made in the spaces provided. Use **sharp pencils** for your drawings. Colored pencils or crayons should **not** be used.

No additional sheets of writing paper are to be inserted in the booklet.

Work on additional sheets will **not** be scored.

ITEM ONE

Daniel owns a small stall where he prepares fruit dishes.Lately, his customers have been asking for fruit slices with added flavor. They prefer slices of Fruit A to be firm and those of Fruit B to be soft.

Daniel has found two available ingredients, A and B,but he is not sure which one will work best to achieve the desired results.He usually slices and prepares the fruits exactly 30 minutes before serving .Now ,he plans to add the ingredient at the same time but does not know which one will produce the expected outcome.

You are provided with fruits **A** and **B**, and ingredients **X** and **Y**.

Task

Design and carry out a scientific investigation to recommend whether Daniel should use ingredient X, ingredient Y, or both to achieve the desired results.

Your design and investigation should include the following

a) aim, hypothesis, variables, apparatus/requirements

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b) i) procedure to determine the effect of ingredients X and Y on fruits A and B

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ii) results of the experiment

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ii) Which ingredient(s) do you recommend Daniel to use to achieve the desired results?

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c) Provide an explanation of your observations to help Daniel understand your recommendation

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ITEM TWO

During a severe drought, a farmer noticed that most of his crops withered and died. However, he observed that a certain unfamiliar **Plant M**, not only survived but also continued to spread across the garden despite the harsh conditions.
When the rains returned, **Plant L**, one of his cultivated crops, began to sprout again. This made the farmer wonder what made these two plants different from the rest.

You are provided with **Specimens L and M**, which are parts of **Plant L** and **Plant M**, respectively.

Task

a) Using observable features, compare **Specimen L** and **Specimen M**.

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b) Explain why **Plant M** was able to survive the dry conditions.

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c) Using a well-labeled drawing, illustrate the key structures of **Specimen L** that enabled it to sprout after the dry season.

FOR SCORE'S USE ONLY

ITEM……..

Basis code

score

ITEM……

Basis code

score